
Autonomous Experimentation Agents for Scientific Discovery

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Literature Survey
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Abstract

Autonomous experimentation agents are an emerging area of Human + AI Scientific Discovery because they can help accelerate scientific research in fields where experiments are expensive, time-consuming, and difficult to scale. In chemistry, biology, drug discovery, and materials science, AI systems can support researchers by generating hypotheses, selecting experiments, optimizing workflows, and analyzing results. However, the current literature is fragmented across self-driving laboratories, machine learning-based planning methods, robotic systems, and LLM-based scientific agents. This survey reviews recent work on autonomous experimentation systems, primarily from 2020–2026, with a focus on chemical and biological domains. We organize the reviewed papers using a taxonomy that compares each system by system type, level of autonomy, methods used, and evaluation approach. Our findings show a shift from isolated AI tools toward more integrated closed-loop systems, where experiment planning, execution, analysis, and iteration are connected in a single pipeline. Bayesian optimization remains a dominant method for sample-efficient experiment planning, while reinforcement learning, active learning, and LLM-based agents are expanding the range of possible workflows. At the same time, major challenges remain, including limited real-world autonomy, difficulty integrating AI decisions with physical laboratories, safety concerns, hallucination, and the need for continued human oversight.

1 Introduction

Autonomous experimentation agents are becoming an increasingly important part of Human + AI Scientific Discovery because they offer a way to accelerate scientific progress beyond the limits of manual experimentation. In fields such as chemistry, biology, drug discovery, and materials science, experiments are often expensive, slow, and difficult to scale [1, 2]. Human researchers must decide which hypotheses to pursue, design experiments, interpret results, and revise their approach based on new evidence. AI systems can support this process by helping generate hypotheses, select promising experiments, optimize workflows, and analyze results more efficiently. More recently, large language models have expanded the role of AI from narrow optimization tools to more flexible scientific agents capable of reasoning over literature, coordinating tools, and assisting with experimental planning [11, 12].

However, the current landscape of autonomous experimentation remains fragmented. Many systems focus on only one part of the discovery pipeline, such as Bayesian optimization, robotic execution, hypothesis generation, or literature interpretation. While prior work has studied self-driving laboratories, machine learning optimization methods, and LLM-based scientific agents separately, there is still a need for a unified synthesis of how these components fit together in end-to-end scientific discovery systems. This is especially important in chemical and biological domains, where real-world laboratory constraints, safety concerns, and human oversight continue to shape what “autonomy” actually means in practice.

This survey addresses the following research questions: (1) How are autonomous experimentation systems designed and structured? (2) What methods are used for experiment planning and optimization? (3) What levels of autonomy are currently achievable? (4) What are the major limitations and open challenges in these systems? The goal is to organize recent work on autonomous experimentation into a clear framework that explains how AI agents, planning algorithms, and laboratory systems interact within modern scientific discovery pipelines.

Our main contributions are:

- We survey fifteen representative autonomous experimentation systems from 2020–2026 across self-driving laboratories, LLM-based scientific agents, and planning-focused systems.
- We develop a taxonomy that compares systems by level of autonomy, core methodology, and experimental setting.
- We analyze how different AI methods, including Bayesian optimization, reinforcement learning, active learning, and LLM-based agents, contribute to scientific discovery workflows.
- We identify key limitations and open challenges, including reliability, hallucination, physical lab integration, safety, and the continued need for human oversight.

The remainder of this paper is organized as follows. Section 2 reviews related work on self-driving laboratories, machine learning-based experiment planning, and LLM-driven scientific agents. Section 3 describes our literature survey methodology. Section 4 presents major findings and trends. Section 5 discusses broader implications, limitations, ethical concerns, and future work. Section 6 concludes the paper.

2 Related Work

Research on autonomous experimentation has developed across several parallel areas, including self-driving laboratories, machine learning-based experiment optimization, and more recently, LLM-driven scientific agents. Prior work has largely focused on individual components of the experimental pipeline rather than fully integrated systems.

Self-driving laboratories. Early work demonstrated closed-loop systems combining robotic platforms with Bayesian optimization for materials discovery and chemical synthesis [2, 3, 1]. These systems showed that automated experiment selection can significantly accelerate discovery, but were typically domain-specific and relied on structured inputs and predefined workflows. The Robot Scientist Adam [5] was an early landmark demonstrating full hypothesis-to-experiment autonomy in a narrowly scoped biological setting.

Machine learning for experiment planning. A large body of research has explored experiment planning methods, particularly Bayesian optimization [17], active learning [16], and reinforcement learning. These approaches focus on sample-efficient optimization in expensive experimental settings, but are often studied in isolation without integration into full laboratory pipelines. Bayesian optimization is especially common because it is well suited to scientific domains where each experiment is costly and only a limited number of trials can be performed.

LLM-based scientific agents. More recent work has introduced LLM-based scientific agents, such as ChemCrow [12] and Coscientist [11], which extend beyond optimization by incorporating hypothesis generation, natural language reasoning, and tool use. These systems represent a shift toward more general-purpose scientific assistants but often lack tight integration with physical lab execution. ResearchAgent [14] and broader multi-agent frameworks [18] further develop this direction by treating scientific discovery as an iterative process involving idea generation, critique, and refinement.

Positioning this survey. Existing surveys tend to focus on either robotic laboratories or machine learning methods independently, without fully addressing how modern systems integrate LLMs, planning algorithms, and experimental hardware into unified architectures. Prior surveys also rarely compare levels of autonomy across systems or analyze the specific role of LLMs in experimental workflows. Unlike prior work, this survey provides a unified view by integrating all three categories

into a single framework, with a specific focus on chemical and biological domains where real-world constraints play a critical role.

3 Methods

This project uses a literature survey methodology to synthesize recent work on autonomous experimentation agents for scientific discovery. Rather than building a new model or dataset, we reviewed existing papers to understand how these systems are designed, what methods they use, how they are evaluated, and what limitations remain.

We selected papers related to autonomous experimentation, self-driving laboratories, machine learning-based experiment planning, and LLM-based scientific agents. Papers were collected through Google Scholar, Semantic Scholar, relevant survey papers, and course materials. Search terms included “autonomous experimentation,” “self-driving laboratory,” “AI for scientific discovery,” “LLM scientific agents,” “Bayesian optimization for chemistry,” “robotic laboratory automation,” “hypothesis generation,” and “active learning in materials science.” In total, we screened approximately 60 papers and included 14 representative systems that met our criteria.

Our search focused mainly on papers from 2020 to 2026 to capture recent developments in LLM-based agents and self-driving laboratories. We also included a few earlier foundational papers when they introduced important ideas such as robotic scientists, closed-loop experimentation, or early autonomous laboratory systems [5, 16]. Papers were included if they: (1) proposed an autonomous or semi-autonomous experimentation system; (2) introduced a planning or optimization method for experimental discovery; (3) applied AI to chemical or biological experimentation; or (4) presented an LLM-based agent for scientific reasoning or experiment design. We excluded papers focused only on scientific writing, summarization, general text processing, or static prediction tasks without a connection to experimental automation.

To organize the survey, we created a taxonomy. Each paper was characterized along four dimensions: system type, level of autonomy, core methods, and evaluation criteria. System types included autonomous experimentation systems, LLM-based agents, planning or hypothesis generation systems, robotic laboratory platforms, and broader survey or framework papers. Levels of autonomy ranged from human-assisted to semi-autonomous to highly autonomous. Methods included Bayesian optimization, reinforcement learning, active learning, robotic automation, LLM-based reasoning, multi-agent systems, and hybrid approaches. Evaluation criteria included experimental efficiency, faster convergence, benchmark performance, hypothesis quality, tool-use accuracy, and qualitative demonstrations.

After organizing the papers, we used thematic comparison to identify major patterns across the literature. We compared systems by structure, methodology, autonomy level, and evaluation approach. This allowed us to identify broad trends, such as the shift toward closed-loop systems, the continued importance of Bayesian optimization, the rise of LLM-based agents, and the remaining challenges of physical lab integration, safety, hallucination, and human oversight.

4 Results

Our review of fifteen representative systems reveals three structural patterns in the autonomous experimentation literature: (1) closed-loop laboratory systems that tightly couple planning, execution, and feedback; (2) LLM-based agent systems that prioritize reasoning, literature integration, and flexible task coordination; and (3) planning-focused systems that contribute specialized optimization or model-discovery methods. Table 1 summarizes the systems across our four taxonomy dimensions.

Shift toward closed-loop integration. Earlier work on self-driving laboratories focused on specific bottlenecks, such as robotic pipetting or Bayesian experiment selection, without connecting these into unified pipelines. More recent systems such as Polybot [3] and the autonomous mobile robotic chemist [8] demonstrate genuinely closed-loop operation: experiment selection, physical execution, measurement, and model updating all occur within a single automated cycle. This architectural shift is significant because the scientific value of any individual component depends critically on how well it interfaces with the others. The agentic AI taxonomy [10] provides a useful formal treatment of

Table 1: Taxonomy of reviewed autonomous experimentation systems. AES = autonomous experimentation system; SDL = self-driving laboratory; BO = Bayesian optimization; RL = reinforcement learning. Autonomy: **H** = high, **M** = medium/semi-autonomous, **L** = low.

System / Paper	Type	Auto.	Core Methods	Evaluation
Mobile Robotic Chemist / Polybot [3]	Core AES	M	BO, active learning, Gaussian processes	Convergence speed and efficiency vs. manual search
Closed-loop superconducting discovery [4]	Core AES	H	BO, surrogate modeling, iterative feedback	Optimal property discovery with fewer experiments
Autonomous mobile robots for synthetic chemistry [8]	Core AES	H	Robotic automation, NMR/LC-MS, heuristic planning	Reaction discovery efficiency and reproducibility
SDL for chemistry & materials [6]	Core AES	M	BO, robotics, ML experiment selection	Throughput and experimental efficiency
SDL for synthetic biology [7]	Core AES	M	Closed-loop experimentation, ML-guided selection	Workflow validation and biological feasibility
Human-aware robots in SDL [9]	Core AES	M	AI perception, intention prediction, robot coordination	Reduced delays in shared laboratory environments
Coscientist [11]	LLM Agent	H	Multi-agent LLMs, tool use, literature retrieval	Hypothesis validity and task completion
ChemCrow [12]	LLM Agent	H	LLMs, chemistry tools, knowledge graphs	Experimental design quality vs. manual workflows
ResearchAgent [14]	LLM Agent	H	Iterative LLM agents, idea generation, experiment design	Novelty and validity of generated research ideas
MoSciBench [13]	Benchmark	H	Multimodal LLM agents, cross-modal reasoning	<50% exact-match accuracy across 88 tasks
BO with experimental constraints [17]	Planning	L	BO, Gaussian-process surrogates	Sample efficiency and optimization quality
RL for computational model discovery [15]	Planning	M	RL policies, computational solvers, reward shaping	Recovery of governing equations
Agentic AI for SDL in soft matter [10]	Framework	M–H	Agent-based modeling, RL concepts, closed-loop systems	Architectural and conceptual analysis
Automating the practice of science [18]	Framework	M–H	AI-driven experiment design, closed-loop learning	Case studies of automated discovery systems

this integration challenge, modeling the lab as an agent-environment system subject to real-world constraints including cost, measurement noise, and safety.

Bayesian optimization as the dominant planning method. Across all three system categories, Bayesian optimization is the most consistently used method for experiment selection. Its prevalence reflects a practical alignment with the structure of scientific discovery: experiments are expensive, evaluations are noisy, and the number of trials must be minimized. Gaussian process surrogates provide principled uncertainty quantification that informs both exploration and exploitation [17]. Reinforcement learning represents a complementary approach for more sequential or dynamic settings, but remains less common in real laboratory deployments due to sample inefficiency and the difficulty of defining appropriate reward functions in practice.

LLM-based agents as flexible scientific coordinators. A rapidly growing cluster of systems uses large language models as the primary decision-making layer. These systems, including ChemCrow [12], Coscientist [11], and ResearchAgent [14], treat LLMs not as narrow optimization functions

but as general-purpose scientific reasoning engines capable of interpreting literature, generating hypotheses, writing experimental protocols, and coordinating domain-specific tools. However, MoSciBench reveals that current LLM agents fall well short of reliable autonomous performance on realistic multimodal scientific tasks, achieving below 50% average accuracy across 88 evaluation instances. The most common failure mode is cross-modal data alignment: the inability to consistently reason jointly over text, tabular data, time series, and molecular structures within a single workflow.

Multimodal and multi-agent architectures. Several systems reflect a move toward architectures that distribute scientific tasks across specialized agents or across multiple data modalities. The autonomous mobile robots for synthetic chemistry [8] exemplify this at the physical level, integrating NMR spectroscopy, LC-MS, and heuristic decision-making within a shared robotic platform. At the computational level, multi-agent frameworks assign roles such as hypothesis generator, critic, and evaluator to different model instances, improving robustness. Despite these advances, cross-modal reasoning remains brittle.

Autonomy is partial in practice. A recurring finding is that “autonomous” in most current systems means semi-autonomous. Even systems classified as high-autonomy depend on human researchers to specify objectives, validate outputs at defined checkpoints, handle edge cases, and intervene when the system produces unsafe or implausible results. This is appropriate given the current reliability of both robotic hardware and LLM-based reasoning, but it complicates direct comparisons across systems, since stated autonomy levels often reflect aspirational framing rather than a precisely defined operational envelope.

5 Discussion

Implications for Human + AI Scientific Discovery. The findings of this survey suggest that autonomous experimentation is most productively understood not as a replacement for human researchers but as a reconfiguration of how human expertise is applied. Rather than running individual experiments, researchers using self-driving laboratories shift toward higher-order tasks: formulating objectives, evaluating system behavior, interpreting surprising results, and making decisions that require domain knowledge beyond the system’s training distribution. This reallocation of human effort is consistent with the broader vision of Human + AI Scientific Discovery as a collaborative framework rather than a fully automated pipeline [18].

At the same time, the results raise questions about where the meaningful bottlenecks in scientific discovery actually lie. Bayesian optimization can reliably reduce the number of experiments needed to identify optimal conditions, but the limiting factor in many domains is not experiment throughput; it is the quality of the hypothesis space being searched. LLM agents can generate hypotheses at scale, but MoSciBench shows that current systems struggle to reason reliably across the heterogeneous data formats that real scientific hypotheses require. Closing this gap may matter more for long-run progress than further improving any individual component of the experimental pipeline.

Limitations of this survey. This survey has several limitations. First, coverage is necessarily incomplete: the literature on autonomous experimentation is growing rapidly, and a comprehensive review of all relevant work from 2020 to 2026 was not feasible within this scope. Papers were selected through keyword-based search rather than systematic bibliometric methods, introducing potential selection bias toward well-cited or widely indexed work. Second, the taxonomy used to organize systems is a simplification. The boundaries between system types are fuzzy, many papers describe systems that combine elements of core autonomous experimentation systems, LLM reasoning, and planning methods, and the autonomy classifications reflect the authors’ descriptions of their systems rather than standardized operational benchmarks. Third, the survey focuses primarily on chemistry and biology, which limits generalizability to other scientific domains such as physics, climate science, or social science, where the structure of experimentation differs substantially.

Threats to validity. The most significant threat to validity is the heterogeneity of evaluation approaches across the reviewed papers. Systems are assessed using different metrics, baselines, task definitions, and experimental settings, making direct quantitative comparison impossible. A second threat is publication bias: the literature disproportionately reports successful systems, while

negative results, failed integrations, unsafe robot behaviors, and hallucinated experimental protocols are systematically underreported.

Ethical considerations. Autonomous experimentation systems raise ethical concerns that deserve explicit attention. In drug discovery and synthetic biology, systems that accelerate beneficial research could also lower barriers to the synthesis of harmful compounds or pathogens. The hallucination problem in LLM-based agents is particularly concerning in safety-critical domains: a system that confidently generates plausible but incorrect experimental protocols could cause physical harm if not carefully validated. Human oversight is not just a technical convenience but an ethical requirement in any domain where experimental outcomes carry risk to researchers, the public, or the environment. Additionally, the concentration of autonomous experimentation capacity in well-resourced institutions may exacerbate existing disparities in scientific access.

Directions for future work. Several directions emerge from this survey. First, the field needs standardized benchmarks for evaluating autonomous experimentation systems across their full operational envelope: not just planning accuracy or task completion in isolation, but integrated performance across hypothesis generation, experiment selection, physical execution, and result interpretation. MoSciBench is a step in this direction, but covers only the computational reasoning component. Second, more work is needed on safe human-AI collaboration protocols in shared laboratory environments, building on the human-aware robot behavior framework [9]. Third, the reliability of LLM-based reasoning in scientific workflows requires systematic evaluation, particularly for hallucination in domain-specific contexts where errors are difficult to detect without expert knowledge. Finally, the field would benefit from more longitudinal studies of self-driving laboratories in sustained real-world use, moving beyond proof-of-concept demonstrations toward evidence about operational robustness and the long-run impact on discovery timelines.

6 Conclusion

This survey has examined the landscape of autonomous experimentation agents for scientific discovery, synthesizing fifteen representative systems from 2020 to 2026 across three categories: closed-loop laboratory systems, LLM-based scientific agents, and planning-focused optimization methods. Our analysis reveals a field in productive transition, moving from isolated AI tools toward more integrated pipelines that connect hypothesis generation, experiment selection, physical execution, and iterative feedback within unified architectures. Bayesian optimization remains the workhorse of experiment planning in chemistry and materials science, while reinforcement learning and active learning are gaining traction in more complex sequential settings. LLM-based agents are expanding the scope of what autonomous systems can reason about, but current performance on realistic multimodal benchmarks falls well short of reliable autonomy. The most consequential open challenge is not any individual methodological gap but the physical-digital integration problem: connecting computational intelligence to real laboratory hardware safely, reliably, and at scale. Autonomous experimentation will not eliminate the need for human scientific expertise; it will redirect it toward the judgment, creativity, and oversight that current systems cannot yet provide.

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